



HUST

TRƯỜNG ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

The background of the slide is a dark blue field filled with a pattern of red dots. These dots are arranged in a way that they form a large, stylized circular shape in the center, with the density of the dots increasing towards the center. The overall effect is a modern, digital, and network-like aesthetic.

SOICT

School of Information and Communication Technology

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Natural Language Processing

1 – Introduction to NLP – Text Representation

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FYI

- *Lecturer: Dr. Bui Thi Mai Anh*
- **Research domains:**
 - **Software Engineering**
 - **Applied AI in Software Engineering**
 - **Search-based Software Engineering:**
evolutionary computing,
reinforcement learning, deep learning
 - **NLP4SE**
- **Lab: Intelligent Software Engineering – BK-AI Center**

<https://iselab-dorlab.github.io/>



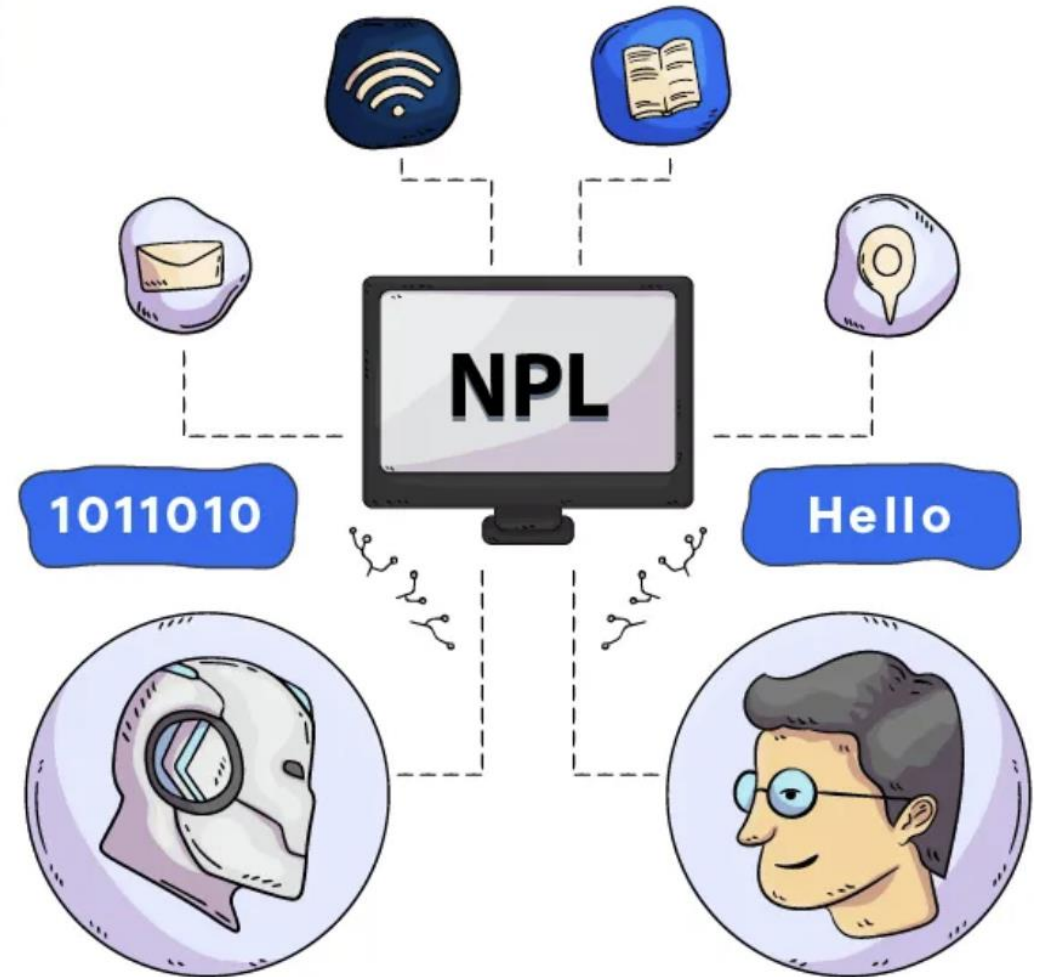


NLP and Applications

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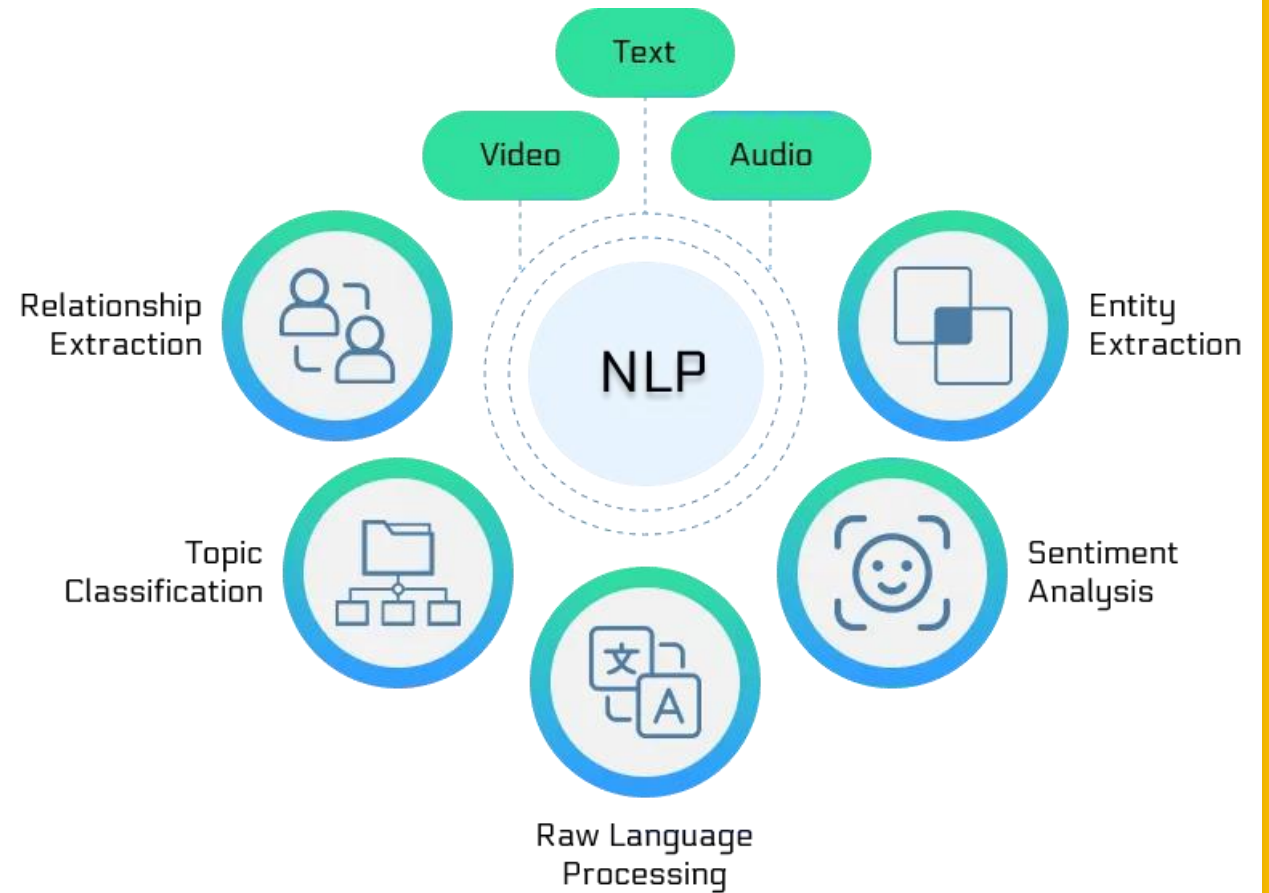
What is NLP?

- The **intersection** of **computer science**, **linguistics**, and **AI**
- Enable machines to **understand**, **interpret**, **generate**, and **respond** to human language



What NLP involves

- **Understanding** language
- **Generating** language
- **Translating** between languages
- **Summarizing** documents
- **Extracting** information
- Sentiment **analysis**



Applications of NLP



Chatbots & Virtual Assistants



Text to Speech



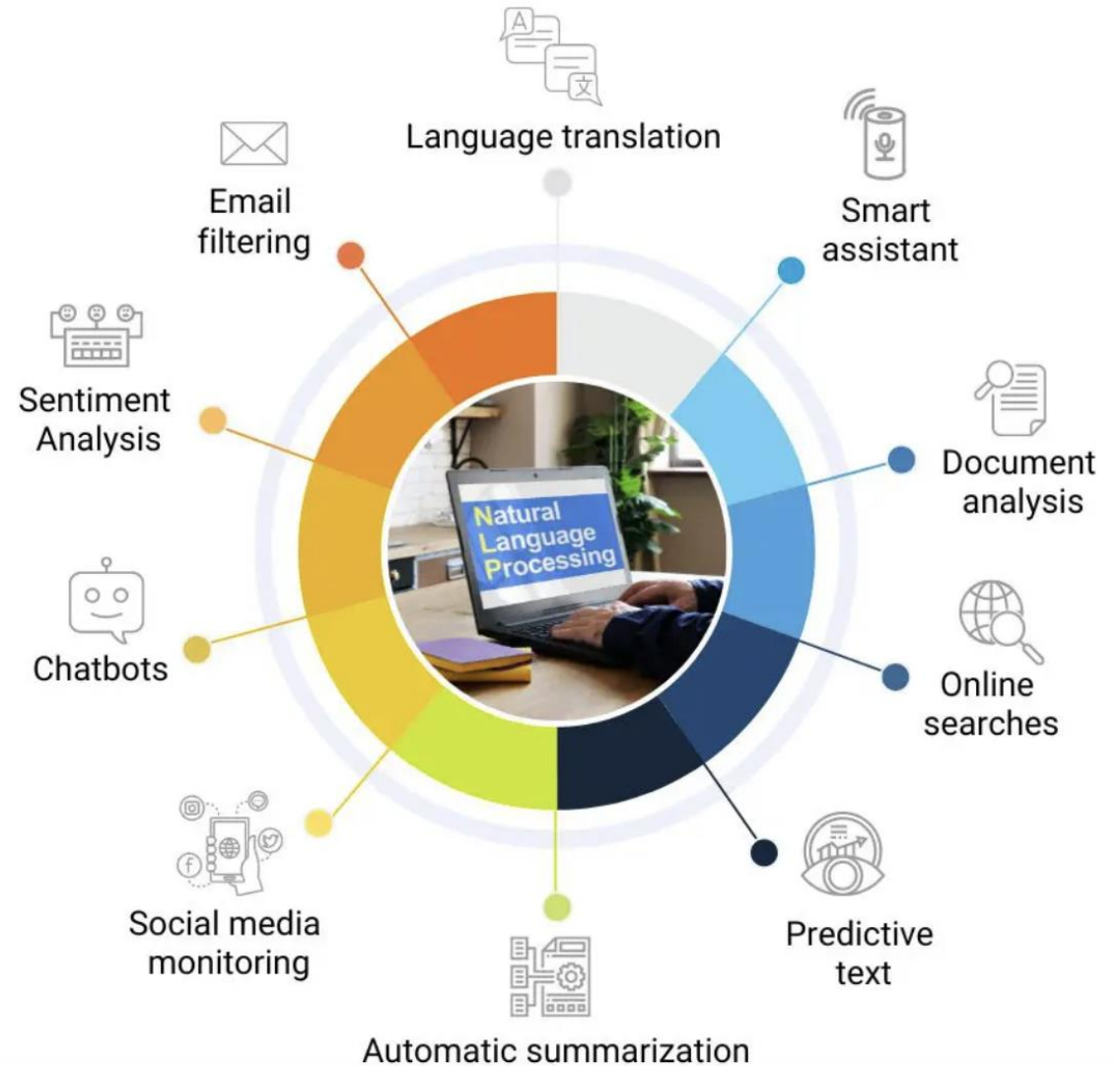
Speech to Text



Sentiment Analysis

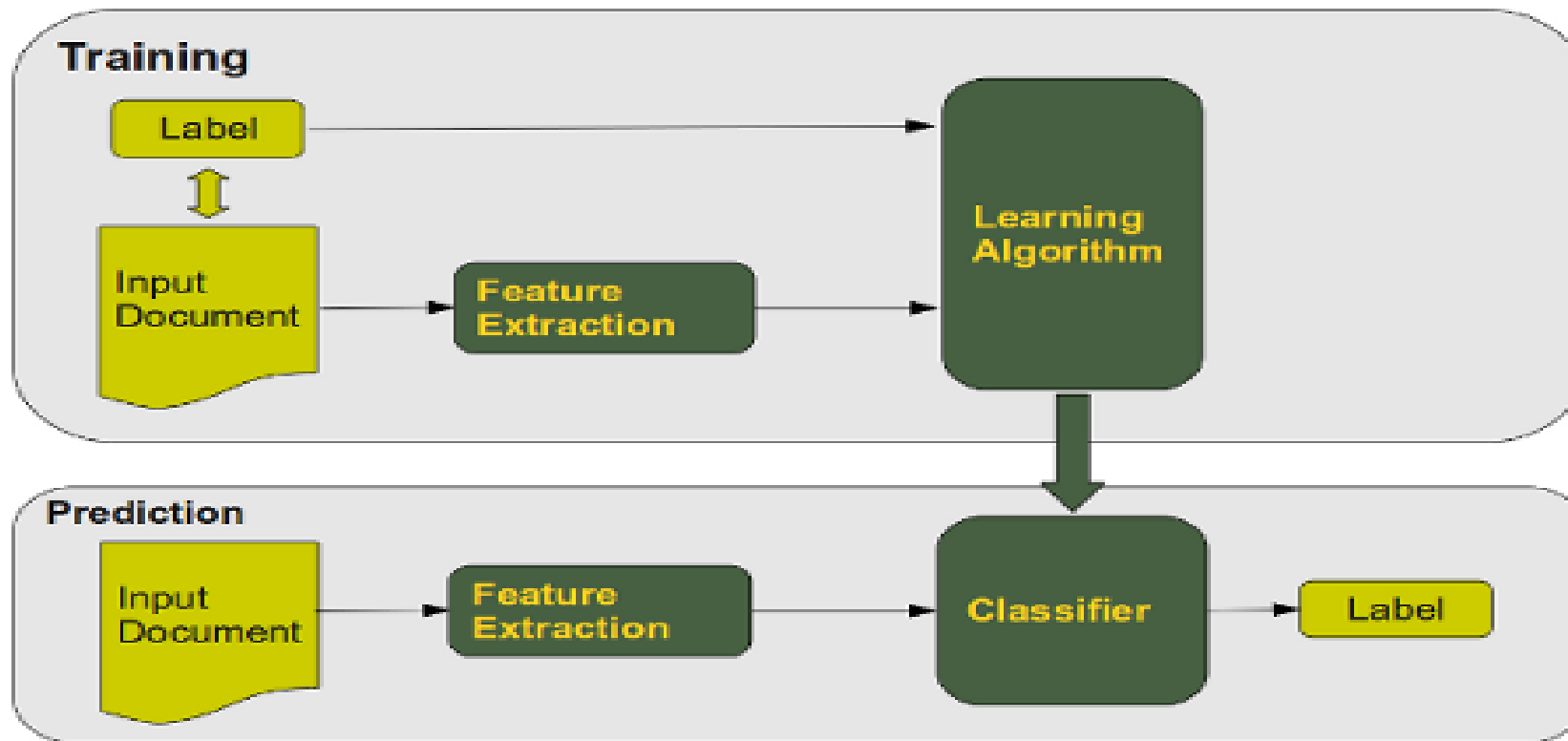


NER



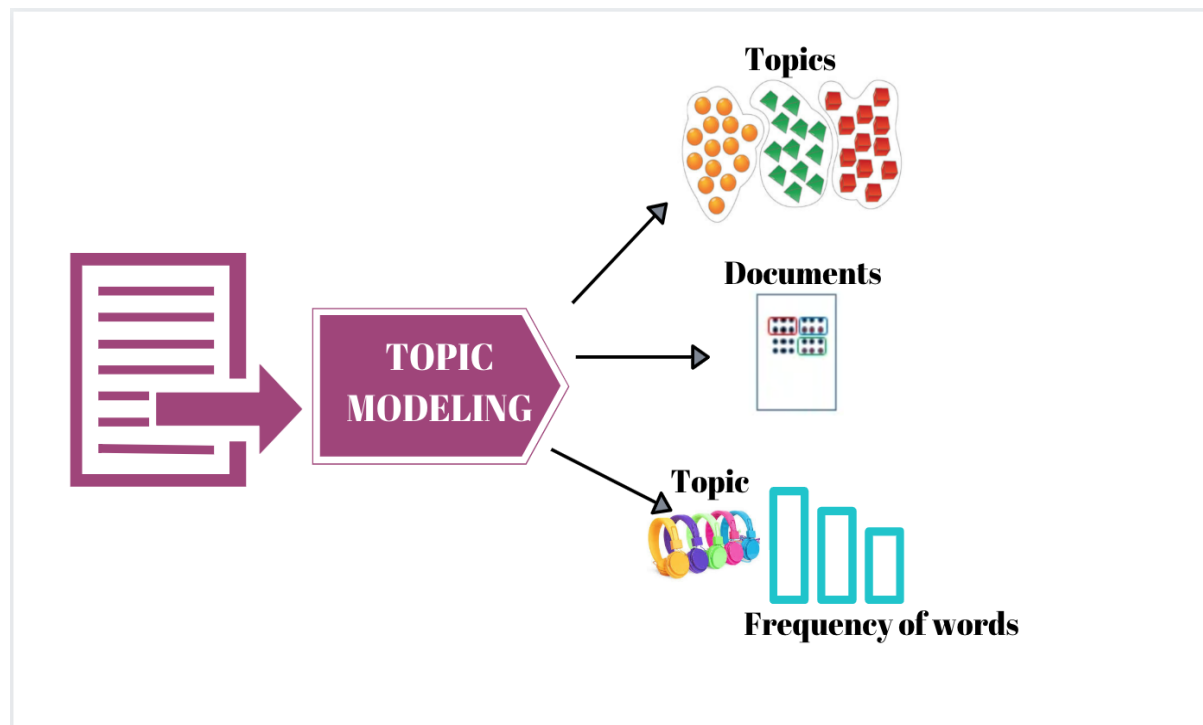
Text Classification

- Classify documents by topic/subject, e.g., spam email detection

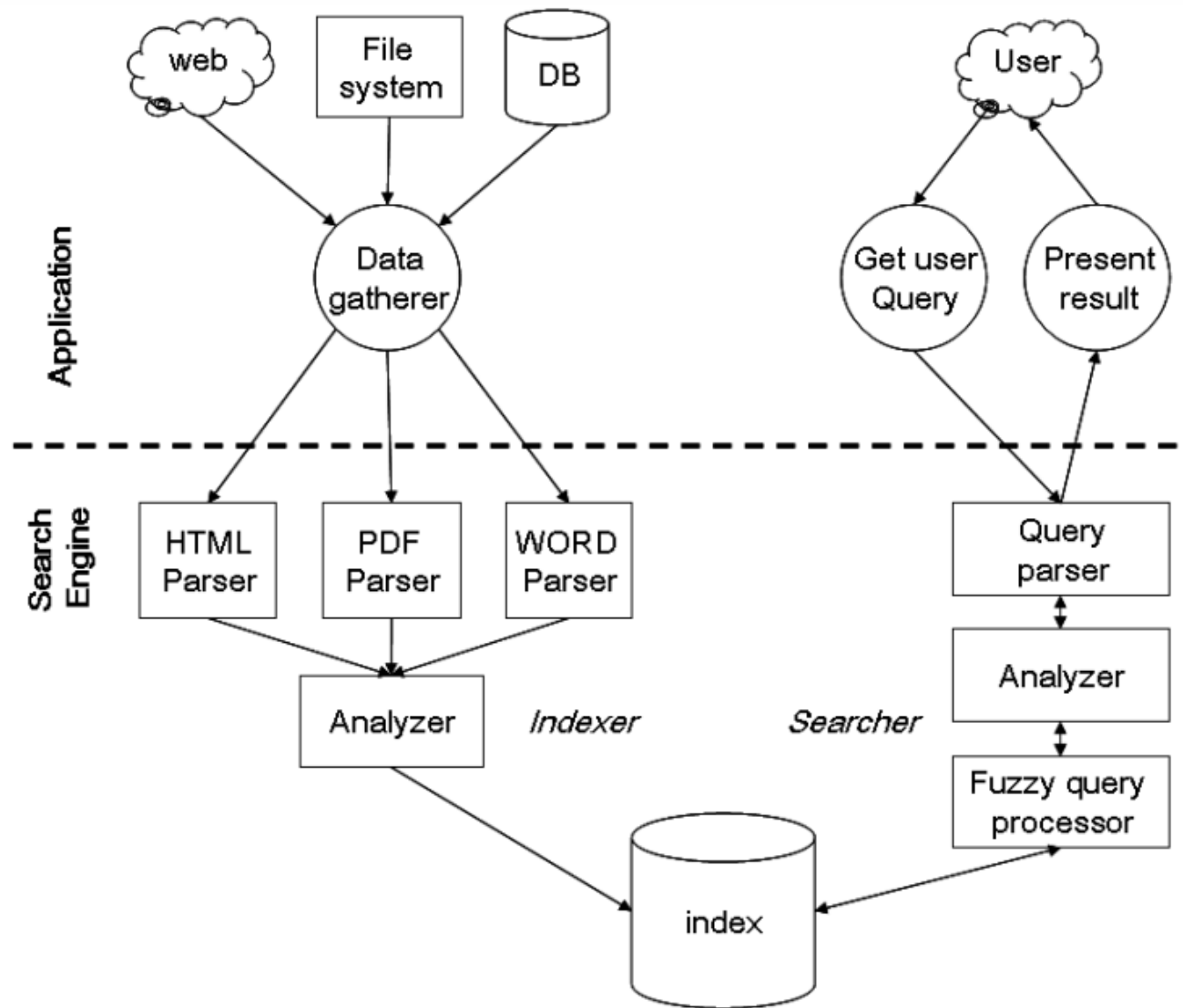


Text Clustering

- From a corpus of documents:
 - What are the **topics**?
 - Clusters of documents?
- Typically, using keywords and frequency of keywords in documents:
 - LSA (Latent Semantic Analysis)
 - LDA (Latent Dirichlet Allocation)
 - NMF (Non-negative Matrix Factorization)



Search Engine



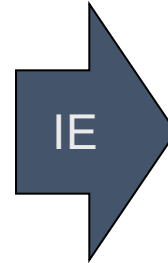
October 14, 2002, 4:00 a.m. PT

For years, [Microsoft Corporation](#) [CEO](#) [Bill Gates](#) railed against the economic philosophy of open-source software with Orwellian fervor, denouncing its communal licensing as a "cancer" that stifled technological innovation.

Today, Microsoft claims to "love" the open-source concept, by which software code is made public to encourage improvement and development by outside programmers. Gates himself says Microsoft will gladly disclose its crown jewels--the coveted code behind the Windows operating system--to select customers.

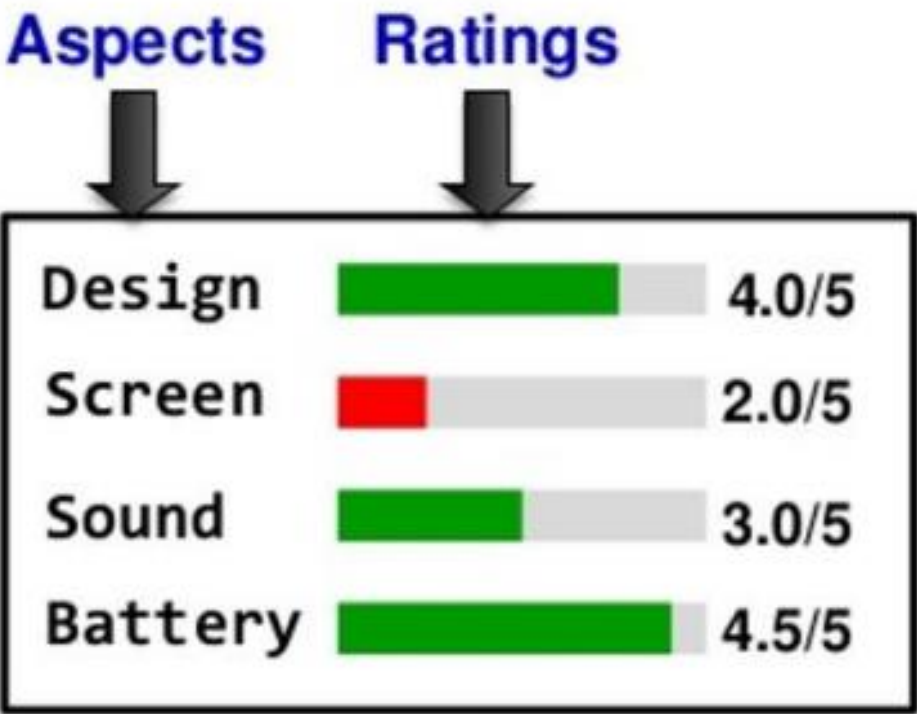
"We can be open source. We love the concept of shared source," said [Bill Veghte](#), a [Microsoft](#) [VP](#). "That's a super-important shift for us in terms of code access."

[Richard Stallman](#), [founder](#) of the [Free Software Foundation](#), countered saying...



NAME	TITLE	ORGANIZATION
Bill Gates	CEO	Microsoft
Bill Veghte	VP	Microsoft
Richard Stallman	founder	Free Soft..

Sentiment Analysis



smartphone xyz

← → ↻ vnexpress-net.translate.google/ha-noi-chap-nhan-dap-an-lam-nham-de-thi-toan-in-mo-4616597.html?_x_tr_sl=vi&_x_tr_tl=en&_x_tr_hl=vi&_x_tr_pto... ⌵ ☆

Google Dịch Việt → Anh ▾ Bản dịch ▾

VNEXPRESS Monday, 12/6/2023 Hanoi ☁ 30° Latest News by region International

Local weather ✕

News View World Video Podcasts business real estate Science Entertainment Sport Law Education Health Life Tourism ≡

Education > Admissions > Common Monday, 12/6/2023, 18:27 (GMT+7)

Hanoi accepts the wrong answer for the math exam

 The Hanoi Department of Education and Training accepts and calculates points for the candidate's wrong answer because the grade 10 math exam is blurry, on the afternoon of June 12.









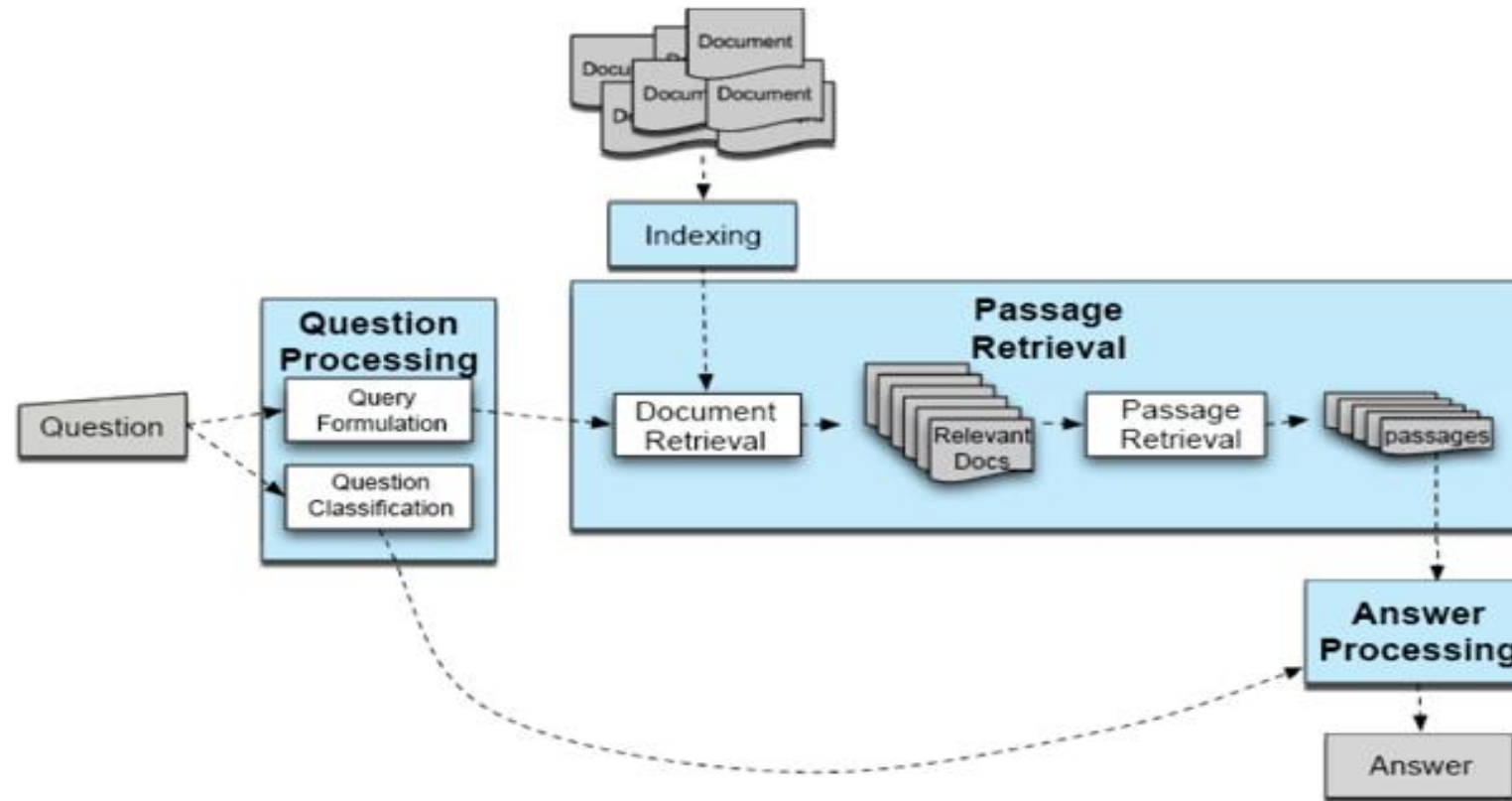
Mr. Ha Xuan Nham, Head of Secondary Education Department, Hanoi Department of Education and Training, said that after reviewing the entire process, comparing the original and the copy, the Department evaluated the math exam questions in grade 10 content errors. However, in the process of copying more than 100,000 copies

BEST SELLING
- Our Most Popular

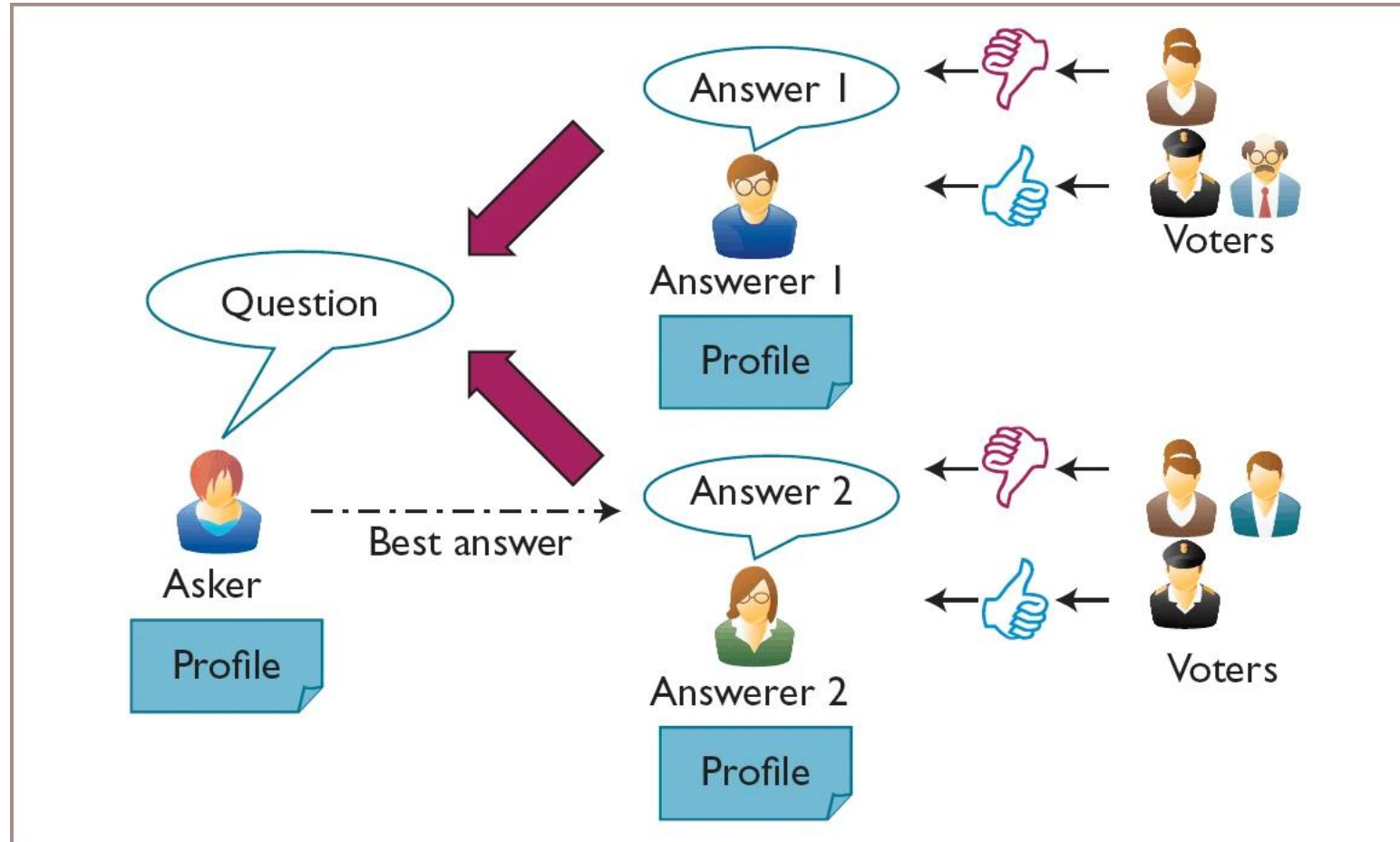
View more

 Answers to the 10th grade English exam in Hanoi

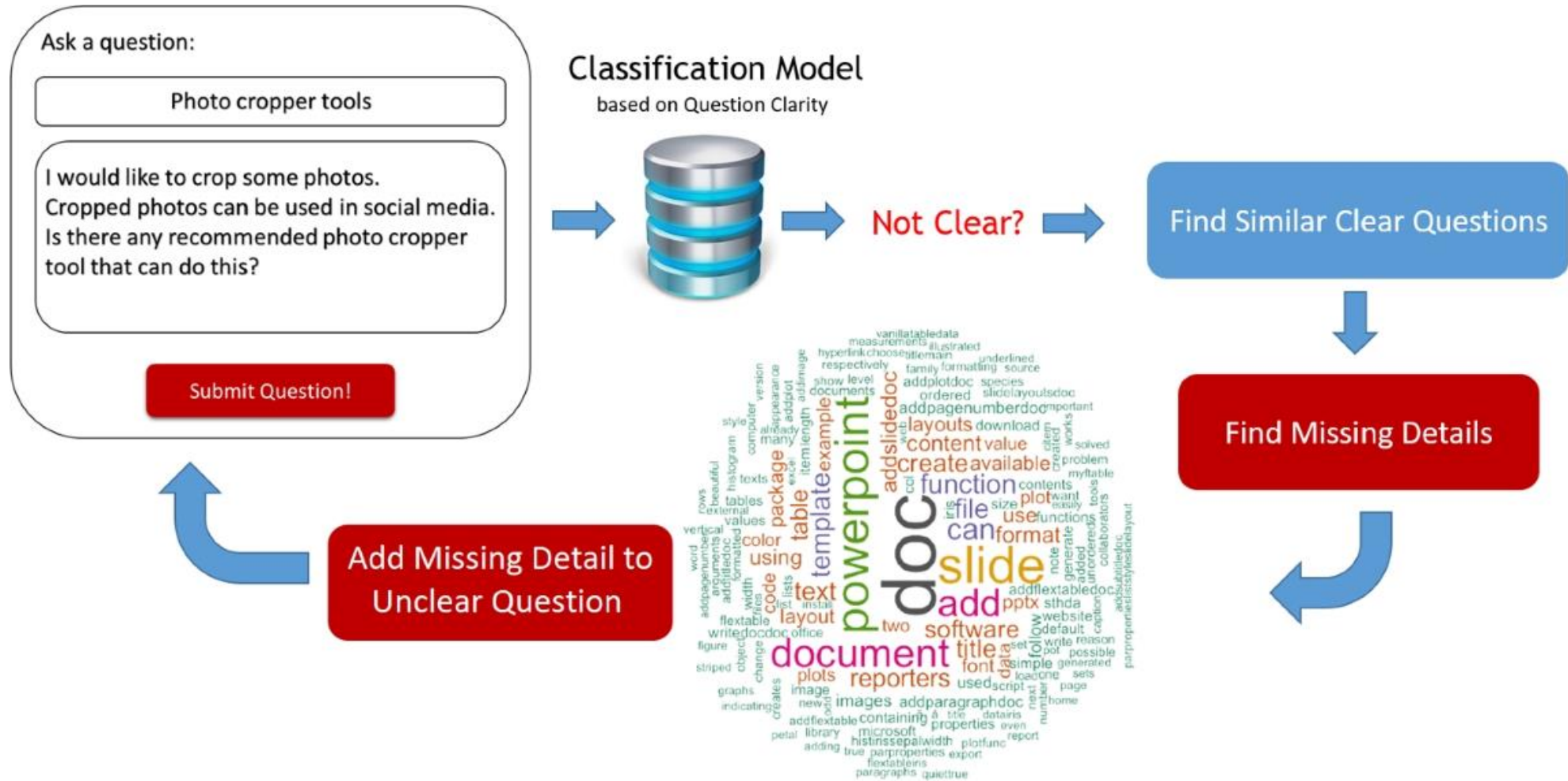
Question Answering – Basic Approach by Information Retrieval



Question Answering – Community Question Answering



Question Answering – Community Question Answering



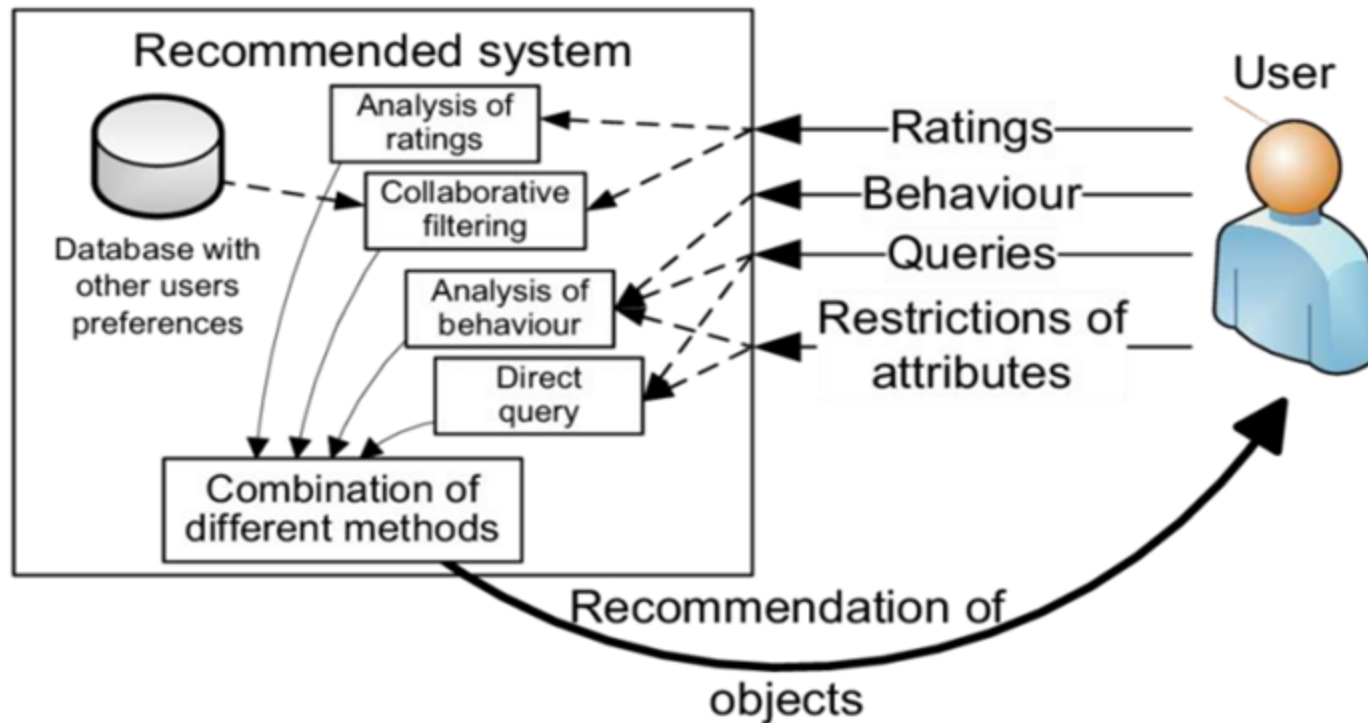
Chatbot – Personal Assistant

Alexa, Google Assistant, Google Dialog-Flow, MS Bot Framework



Recommendation System

Big Ecommerce Platform: Amazon, Alibaba, ...

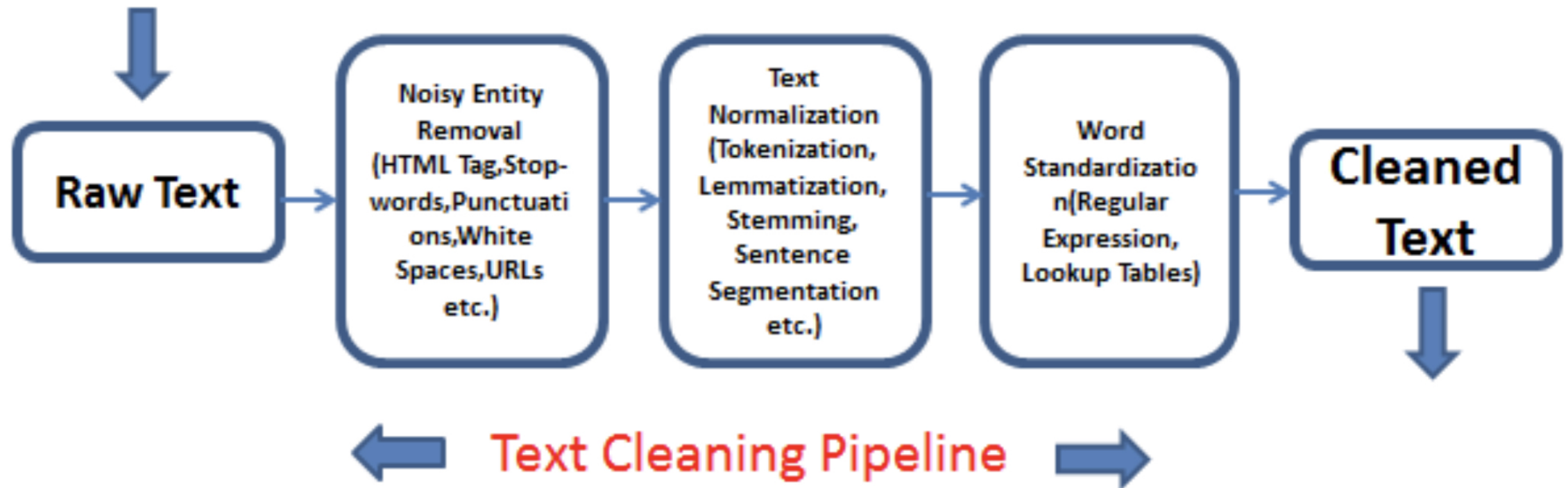




Text Data Pre-Processing

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Pipeline



Uncapitalizing

- Problem: “*NLP*” and “*nlp*” is different or not?
- All the text needs to be represented in the same format
- Solution: using `lower()` function in `python`

Lệnh trừng phạt của Mỹ lên Huawei không chỉ tác động đến các công ty công nghệ Trung Quốc mà còn kéo theo nhiều hệ lụy tới ngành công nghiệp toàn cầu.



lệnh trừng phạt của mỹ lên huawei không chỉ tác động đến các công ty công nghệ trung quốc mà còn kéo theo nhiều hệ lụy tới ngành công nghiệp toàn cầu

Removing Punctuation

- *Punctuation has no meaning!*
- Reducing data dimension as well as improving the computational performance
- Solution: Regular Expression + `replace()` function in python

	A	B	C	D
1	Text with Punctuation			Remove Punctuation
2	"Apple"			Apple
3	(Pear). 5			Pear 5
4	{[Orange]}			Orange
5	Lemon;;; :::			Lemon
6	Lychee!			Lychee
7	<Blueberry>			Blueberry
8	Dash-test			Dashtest
9	TEST~!#\$%^&*()_+{}[]"'";<>?.,			TEST



Removing Stop words

- Stop words are commonly used in text document but **meaningless**
- Removing stop words allows to reduce the data dimension and to improve the model performance
- Solution:
 - Using the library NLTK
 - Building a list of stop-words and removing them from the input document
 - Example: <https://github.com/stopwords/vietnamese-stopwords>

Ngay cả khi trời mưa, trận đấu vẫn diễn ra



Removing stop words

Trời mưa, trận đấu diễn ra

Standardizing Text

- Transforming acronym and abbreviation in the document text
- Solution:
 - Building a dictionary for common acronym and abbreviation in the text

Raw	Normalized
2moro 2mrrw 2morrow 2mrw tomrw	tomorrow
b4	before
otw	on the way
:) :-) ;-)	smile

Correcting Spelling

studing → studying
intresting → interesting
aquire → acquire

- Data comes from social network (e.g., user comments, blogs, tweets etc.) may contain spelling errors
- Correcting spelling allows to remove redundant words
- Solution:
 - Using library TextBlod

Tokenizing

- Motivation: split the input text into “terms” → numerical representation
- Input: document after some pre-processing techniques
- Output: A list of terms
- Solution
 - For English documents: NLTK, TextBlod, Spacy
 - For Vietnamese documents: VnCoreNLP, underthesea, coccoc-tokenizer

VnCoreNLP: <https://github.com/vncorenlp/VnCoreNLP>

Underthesea: <https://github.com/undertheseanlp/underthesea>

Coccoc-tokenizer: <https://github.com/coccoc/coccoc-tokenizer>

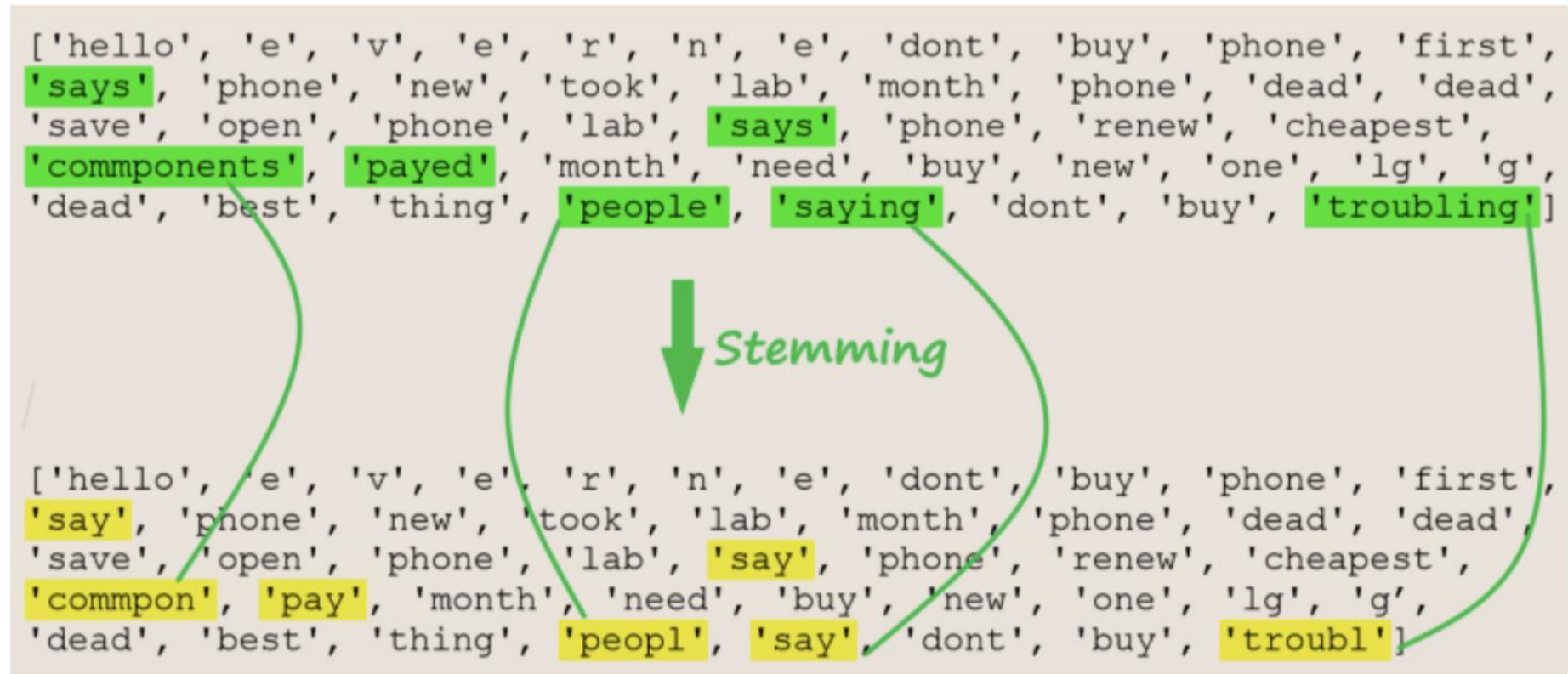
```
'hello e v e r y o n e           dont buy this phone at all first o  
f all that says the phone in new   i took it to the lab after  month the  
phone is dead dead  you can save it they open the phone in the lab and say  
s      the phone is renew  and its cheapest commponents i payed      for on  
ly  month now i need to buy new one this lg g  is dead not a best thing  
people are saying to me dont buy from  at all  it s troubling  '
```



```
['hello', 'e', 'v', 'e', 'r', 'y', 'o', 'n', 'e', 'dont', 'buy', 'this',  
'phone', 'at', 'all', 'first', 'of', 'all', 'that', 'says', 'the',  
'phone', 'in', 'new', 'i', 'took', 'it', 'to', 'the', 'lab', 'after',  
'month', 'the', 'phone', 'is', 'dead', 'dead', 'you', 'can', 'save', 'it',  
'they', 'open', 'the', 'phone', 'in', 'the', 'lab', 'and', 'says', 'the',  
'phone', 'is', 'renew', 'and', 'its', 'cheapest', 'commponents', 'i',  
'payed', 'for', 'only', 'month', 'now', 'i', 'need', 'to', 'buy', 'new',  
'one', 'this', 'lg', 'g', 'is', 'dead', 'not', 'a', 'best', 'thing',  
'people', 'are', 'saying', 'to', 'me', 'dont', 'buy', 'from', 'at', 'all',  
'it', 's', 'troubling']
```

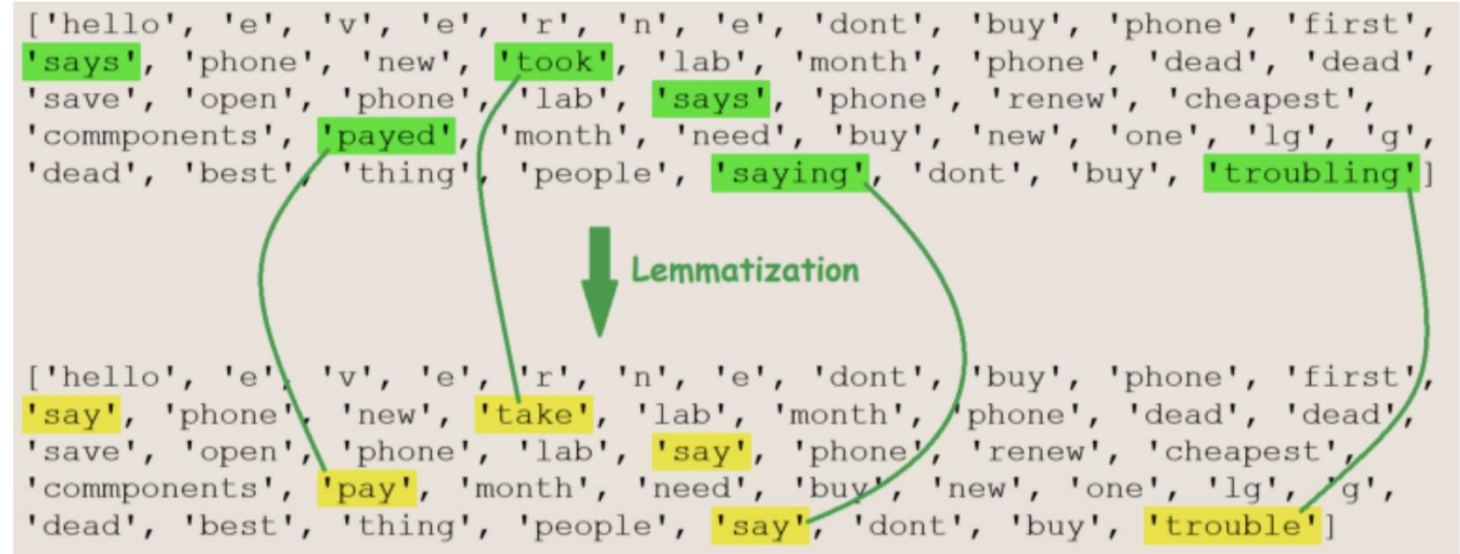
Stemming

- Extract the base form of a word by removing affixes from them
- Solution:
 - NLTK
 - TextBlod



Lemmatization

- Identify the derived forms of a word then convert them to the base form
- Input: a word
- Output: base form of this word
- Solution
 - NLTK
 - TextBlod



-
- A word cloud of business-related terms in various sizes and orientations. The most prominent words are 'STRATEGY', 'VISION', 'TEAMWORK', 'MARKETING', 'SUCCESS', 'LEADERSHIP', 'PLANNING', 'DEVELOPMENT', 'COACHING', 'SOLUTION', 'PROMOTE', 'IDEAS', 'GOALS', 'TARGET', 'ANALYSIS', 'RELATION', 'BUSINESS', 'MEDIA', 'POWER', 'CREATE', 'ACHIEVE', 'STEPS', 'COMMUNICATION', 'LEARN', 'PROCESS', 'IDEAS', 'Creation', 'Business', 'Team', 'sepi', 'ideas', 'POWER', 'ideas', 'Solution', 'POWER', 'IDEAS', 'analysis', 'Create', 'analysis', 'BUSINESS', 'STRATEGY', 'PLANNING', 'communication', 'SUCCESS', 'PROMOTE', 'IDEAS', 'by Getty Images'.



Information Extraction from text data

What is information extraction?

- Automatically extract structured information from unstructured information
- Re-organize information in a systematic way
- Extracted information is stored in database for further usage

What is information extraction?

October 14, 2002, 4:00 a.m. PT

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NAME	TITLE	ORGANIZATION
Bill Gates	CEO	Microsoft
Bill Veghte	VP	Microsoft
Richard Stallman	founder	Free Soft..

Closed set

U.S. states

He was born in Alabama...

The big Wyoming sky...

Complex pattern

U.S. postal addresses

University of Arkansas

P.O. Box 140

Hope, AR Headquarters:

1128 Main Street, 4th Floor

Cincinnati, Ohio 45210

Regular set

U.S. phone numbers

Phone: (413) 545-1323

The CALD main office is 412-268-1299

Ambiguous patterns, needing context and many sources of evidence

Person names

...was among the six houses sold
by Hope Feldman that year.

Pawel Opalinski, Software
Engineer at WhizBang Labs.

Named Entity Extraction

Jack Welch will retire as CEO of General Electric tomorrow. The top role at the Connecticut company will be filled by Jeffrey Immelt.

Single entity

Person: Jack Welch

Person: Jeffrey Immelt

Location: Connecticut

Binary relationship

Relation: Person-Title

Person: Jack Welch

Title: CEO

Relation: Company-Location

Company: General Electric

Location: Connecticut

N-ary record

Relation: Succession

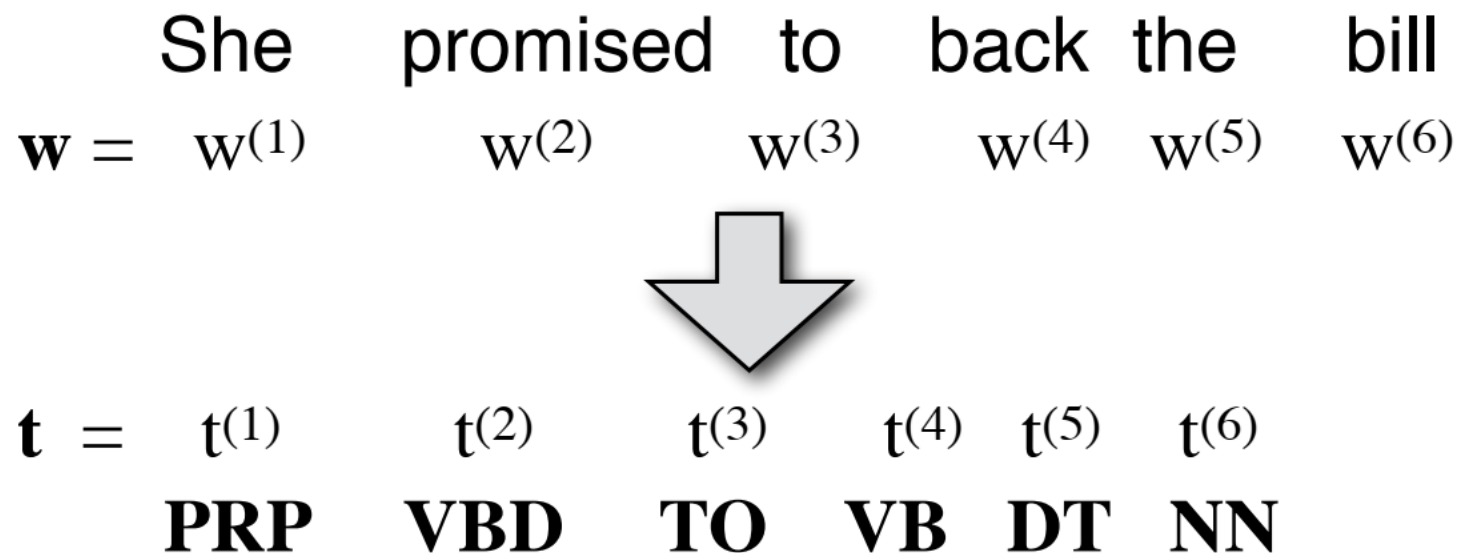
Company: General Electric

Title: CEO

Out: Jack Welsh

In: Jeffrey Immelt

Sequence Labeling



- Given a sequence input $\mathbf{w}=w^{(1)}...w^{(n)}$, predict the best-matched sequence of tags (tagging) $\mathbf{t}=t^{(1)}...t^{(n)}$

$$\mathbf{t}^* = \operatorname{argmax}_{\mathbf{t}} P(\mathbf{t} \mid \mathbf{w})$$

English Pos-tagging

- Brown corpus: 87 tags
- 3 benchmark corpus:
 - Small: 45 tags - Penn treebank
 - Medium: 61 tags, British national corpus
 - Large: 146 tags, C7

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	<i>and, but, or</i>	SYM	Symbol	<i>+, %, &</i>
CD	Cardinal number	<i>one, two, three</i>	TO	“to”	<i>to</i>
DT	Determiner	<i>a, the</i>	UH	Interjection	<i>ah, oops</i>
EX	Existential ‘there’	<i>there</i>	VB	Verb, base form	<i>eat</i>
FW	Foreign word	<i>mea culpa</i>	VBD	Verb, past tense	<i>ate</i>
IN	Preposition/sub-conj	<i>of, in, by</i>	VBG	Verb, gerund	<i>eating</i>
JJ	Adjective	<i>yellow</i>	VCN	Verb, past participle	<i>eaten</i>
JJR	Adj., comparative	<i>bigger</i>	VBP	Verb, non-3sg pres	<i>eat</i>
JJS	Adj., superlative	<i>wildest</i>	VBZ	Verb, 3sg pres	<i>eats</i>
LS	List item marker	<i>1, 2, One</i>	WDT	Wh-determiner	<i>which, that</i>
MD	Modal	<i>can, should</i>	WP	Wh-pronoun	<i>what, who</i>
NN	Noun, sing. or mass	<i>llama</i>	WP\$	Possessive wh-	<i>whose</i>
NNS	Noun, plural	<i>llamas</i>	WRB	Wh-adverb	<i>how, where</i>
NNP	Proper noun, singular	<i>IBM</i>	\$	Dollar sign	<i>\$</i>
NNPS	Proper noun, plural	<i>Carolinas</i>	#	Pound sign	<i>#</i>
PDT	Predeterminer	<i>all, both</i>	“	Left quote	<i>‘ or “</i>
POS	Possessive ending	<i>'s</i>	”	Right quote	<i>’ or ”</i>
PRP	Personal pronoun	<i>I, you, he</i>	(	Left parenthesis	<i>[, (, {, <</i>
PRP\$	Possessive pronoun	<i>your, one's</i>	)	Right parenthesis	<i>],), }, ></i>
RB	Adverb	<i>quickly, never</i>	,	Comma	<i>,</i>
RBR	Adverb, comparative	<i>faster</i>	.	Sentence-final punc	<i>. ! ?</i>
RBS	Adverb, superlative	<i>fastest</i>	:	Mid-sentence punc	<i>: ; ... --</i>
RP	Particle	<i>up, off</i>			

Vietnamese Pos-Tagging

Câu tiếng Việt đã tách từ	Qua những lần từ Sài_Gòn về Quảng_Ngãi kiểm_tra công_việc , Sophie và Jane thường trò_chuyện với Mai , cảm_nhận ngọn_lửa_sống và niềm_tin mãnh_liệt từ người phụ_nữ VN này .															
Câu tiếng Việt đã được gán nhãn từ loại	Qua những lần từ Sài_Gòn về Quảng_Ngãi kiểm_tra công_việc , Sophie và Jane thường trò_chuyện với Mai , cảm_nhận ngọn_lửa_sống và niềm_tin mãnh_liệt từ người phụ_nữ VN này .															
Chú thích từ loại	<table><tr><td>DANH TỪ</td><td>SỐ TỪ</td><td>THÁN TỪ</td></tr><tr><td>ĐỘNG TỪ</td><td>PHỤ TỪ</td><td>TRỢ TỪ</td></tr><tr><td>TÍNH TỪ</td><td>GIỚI TỪ</td><td>TỪ ĐƠN LẺ</td></tr><tr><td>ĐẠI TỪ</td><td>CẢM TỪ</td><td>TỪ VIẾT TẮT</td></tr><tr><td>ĐỊNH TỪ</td><td>LIÊN TỪ</td><td>KHÔNG XÁC ĐỊNH</td></tr></table>	DANH TỪ	SỐ TỪ	THÁN TỪ	ĐỘNG TỪ	PHỤ TỪ	TRỢ TỪ	TÍNH TỪ	GIỚI TỪ	TỪ ĐƠN LẺ	ĐẠI TỪ	CẢM TỪ	TỪ VIẾT TẮT	ĐỊNH TỪ	LIÊN TỪ	KHÔNG XÁC ĐỊNH
DANH TỪ	SỐ TỪ	THÁN TỪ														
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ĐẠI TỪ	CẢM TỪ	TỪ VIẾT TẮT														
ĐỊNH TỪ	LIÊN TỪ	KHÔNG XÁC ĐỊNH														

Named Entity Recognition

When **Sebastian Thrun** PERSON started at **Google** ORG in **2007** DATE, few people outside of the company took him seriously. “I can tell you very senior CEOs of major **American** NORP car companies would shake my hand and turn away because I wasn’t worth talking to,” said **Thrun** PERSON, now the co-founder and CEO of online higher education startup Udacity, in an interview with **Recode** ORG **earlier this week** DATE.

A little **less than a decade later** DATE, dozens of self-driving startups have cropped up while automakers around the world clamor, wallet in hand, to secure their place in the fast-moving world of fully automated transportation.

- Thực thể (entity) là đối tượng hoặc tập hợp các đối tượng trong thế giới tự nhiên được mô tả bằng ngôn ngữ

- Phân loại:

- Tên người
- Tên địa điểm
- Tên tổ chức
- Giá trị số
- Thời gian

Pierre Vinken , 61 years old , will join IBM 's board
as a nonexecutive director Nov. 29 .



[PERS Pierre Vinken] , 61 years old , will join
[ORG IBM] 's board as a nonexecutive director
[DATE Nov. 2] .

Approaches

Rule-based

- Email, Time, Phone Number, URL

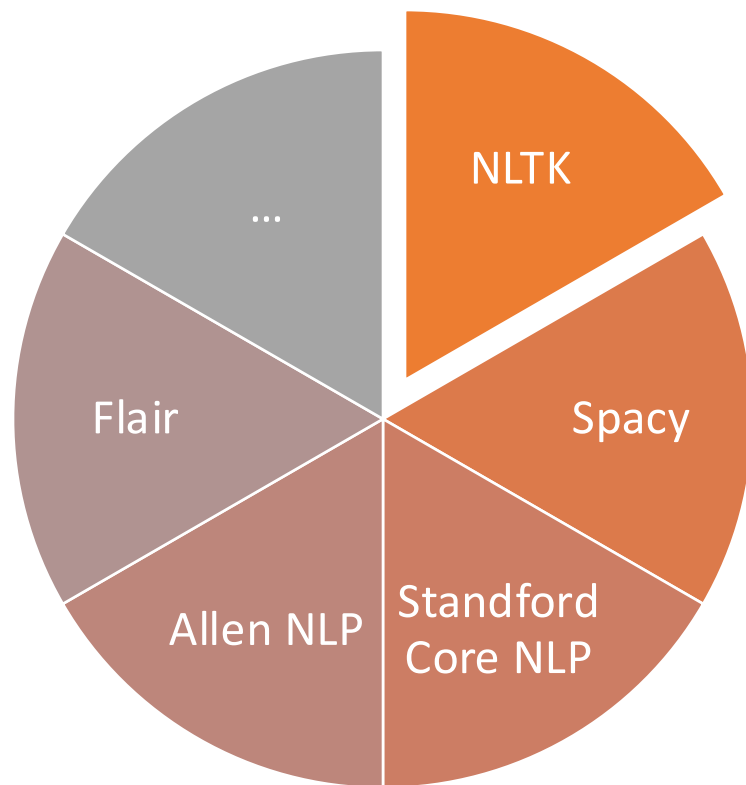
Machine learning

- Hidden Markov Model - HMM
- Maximum Entropy Markov Model (MEMM)
- Conditional Random Field (CRF)

Deep Learning

- RNN/LSTM
- BERT

Libraries





Text Representation

The meaning of a word

- Meaning:
 - the idea that is represented by a word, phrase, etc.
 - the idea that a person wants to express using words, phrases, etc.
 - the idea that is expressed in a work of writing, art, etc.

signifier (symbol) $\Leftrightarrow$ signified (idea or thing)

= denotational semantics

tree $\Leftrightarrow$ { , , , ... }

Overview of Text Representation Development

- Text needs to be represented as numbers/vectors so that computer programs can process them
- How were text represented in history?

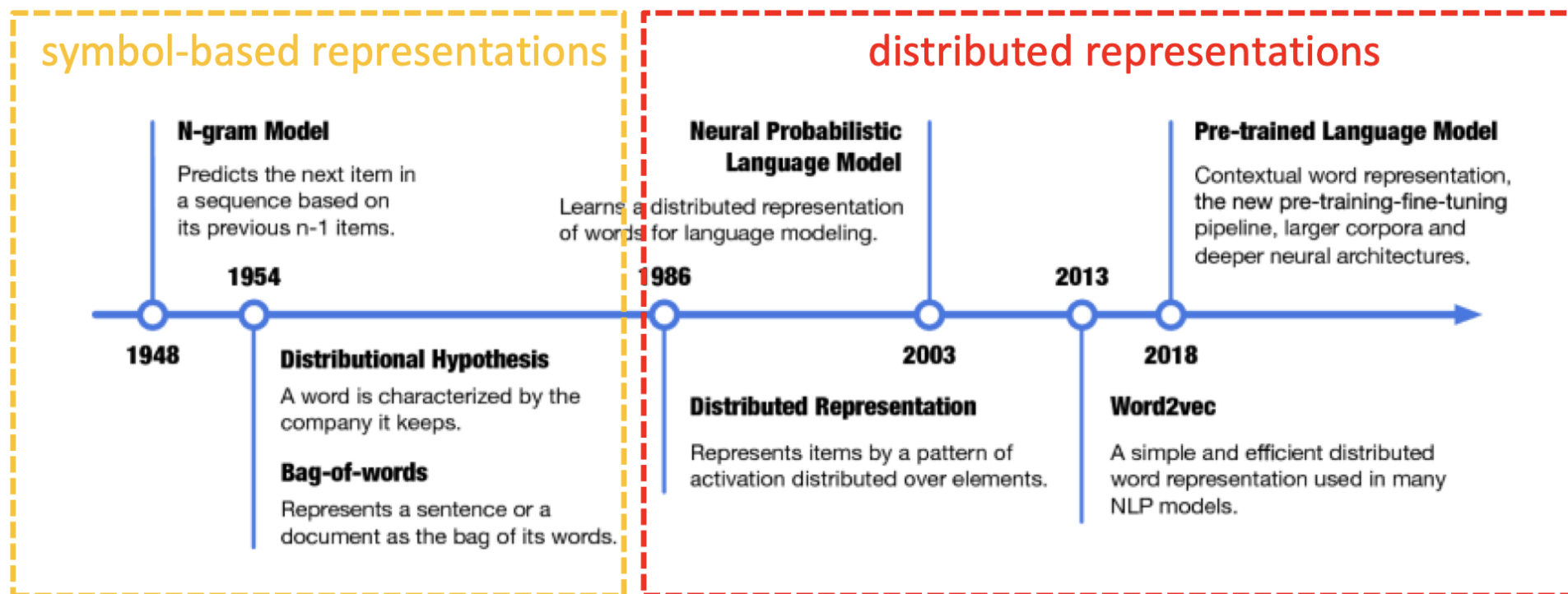


Figure from: Liu Z., Lin Y., Sun M. (2020) Representation Learning and NLP. In: Representation Learning for Natural Language Processing. Springer, Singapore.

Symbol-based Text Representation

- a.k.a Discrete Vector Representation
- One-hot encoding vector
- Bag-of-words model
- TF-IDF
- Co-occurrence matrix

One-hot encoding

- Each word w in the corpus library is given a unique integer that is between 1 and $|V|$ (V : vocabulary)
- Each word is then represented by a $|V|$ dimensional binary vector of 0s and 1

```
#function to return one-hot representation of passed text
def get_onehot_representation(text, vocab = data_vocab):
    onehot_encoded = []
    for word in text:
        temp = [0]*len(vocab)
        temp[vocab.index(word)-1] = 1
        onehot_encoded.append(temp)
    return onehot_encoded

print("One Hot Representation for sentence \"the cat sat on the mat\" :)")
get_onehot_representation(['the', 'cat', 'sat', 'on', 'the', 'mat'], sample_vocab)
```

The cat sat on the mat
Main Notebook Content

The: [0 1 0 0 0 0 0]

cat: [0 0 1 0 0 0 0]

sat: [0 0 0 1 0 0 0]

on: [0 0 0 0 1 0 0]

the: [0 0 0 0 0 1 0]

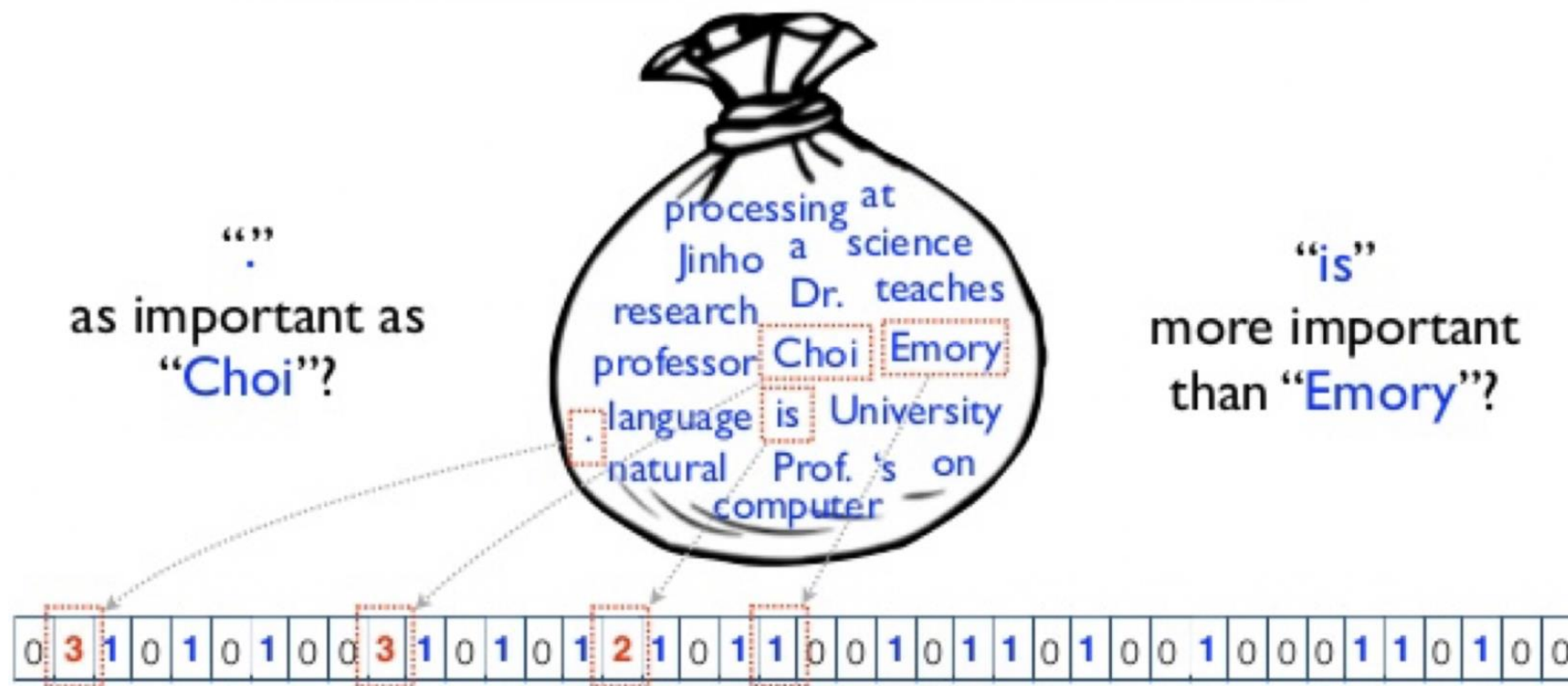
mat: [0 0 0 0 0 0 1]

Drawbacks

- The size of a one-hot vector is directly proportional to the size of the vocabulary → **high-dimensional** vector, very **sparse**!
- One-hot representation does **not** give a **fixed-length** representation for text, i.e., the sentence with 32 words in it and 40 words in it has variable length representation
- One-Hot representation gives each word the **same weight**
- One-Hot representation does **not represent the meaning** of the word
- Encounter **Out-Of-Vocabulary** (OOV) problem

Bag of words (BOW)

Jinho Choi is a professor at Emory University .
Prof. Choi teaches computer science .
Dr. Choi's research is on natural language processing .



- Represent the text under consideration as a bag (collection) of words while **ignoring** the **order** and **context**.
- A document is represented as an integer vector of $|V|$ dimensions
- The i^{th} element = the number of time the i^{th} word appears in the document

Vocabulary = {the, cat, sat, in, hat, with}

examples

Document	the	cat	sat	in	hat	with
<i>the cat sat</i>	1	1	1	0	0	0
<i>the cat sat in the hat</i>	2	1	1	1	1	0
<i>the cat with the hat</i>	2	1	0	0	1	1

Having fixed-length for any sentences!

BOW – Code example

```
from sklearn.feature_extraction.text import CountVectorizer

sample_bow = CountVectorizer()

# sample_corpus = [['the', 'cat', 'sat'],
#                  ['the', 'cat', 'sat', 'in', 'the', 'hat'],
#                  ['the', 'cat', 'with', 'the', 'hat']]

sample_corpus = ["the cat sat", "the cat sat in the hat", "the cat with the hat"]

sample_bow.fit(sample_corpus)

def get_bow_representation(text):
    return sample_bow.transform(text)

print(f"Vocabulary mapping for given sample corpus : \n {sample_bow.vocabulary_}")
print("\nBag of word Representation of sentence 'the cat cat sat in the hat'")
print(get_bow_representation(["the cat cat sat in the hat"]).toarray())
```

Vocabulary mapping for given sample corpus :
{'the': 4, 'cat': 0, 'sat': 3, 'in': 2, 'hat': 1, 'with': 5}

Bag of word Representation of sentence 'the cat cat sat in the hat'
[[2 1 1 1 2 0]]

Advantages and Disadvantages

- Simple to implement
- Document having the same words will have their vector representations closer to each other in Euclidean space

S1 = "cat on the mat" --> BoW Representation --> {0 1 1 0 1 0 1}

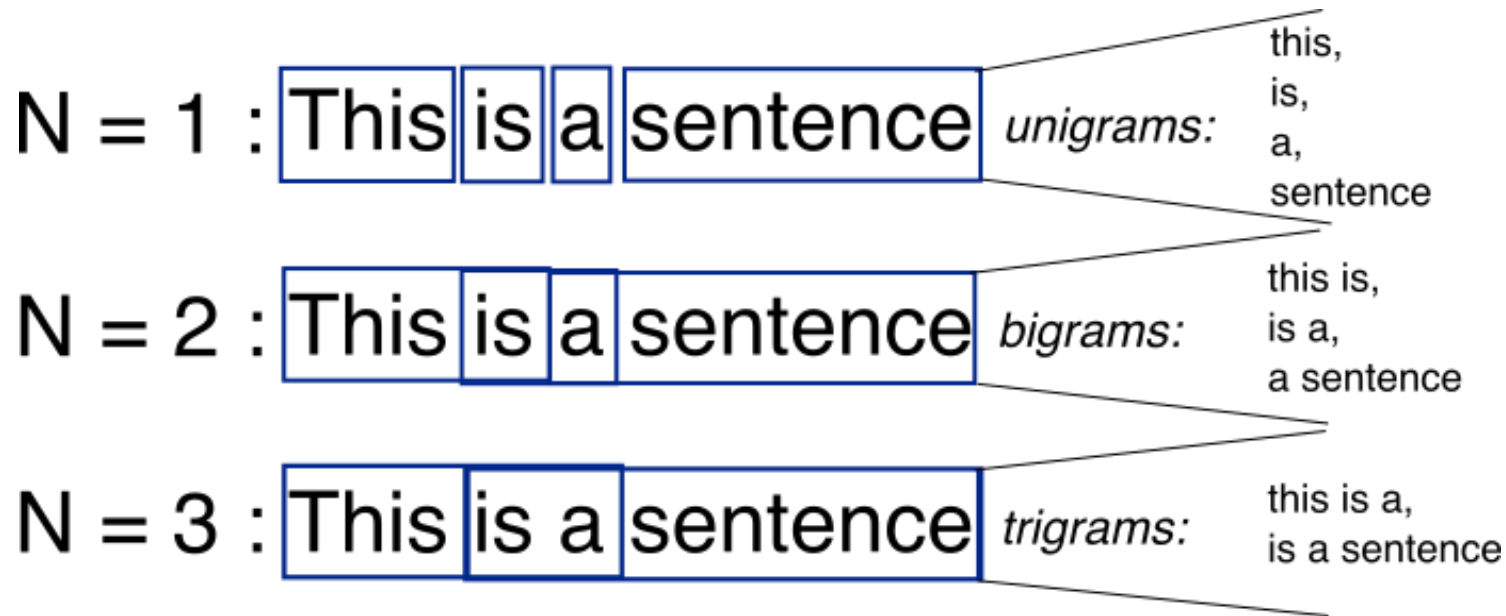
S2 = "mat on the cat" --> BoW Representation --> {0 1 1 0 1 0 1}

S3 = "dog in the mat" --> BoW Representation --> {0 1 0 1 1 1 0}

- Fix-length vector
- High-dimensional vector
- Does not consider the order of words (S1 and S2 having the same representation)
- OOV problem

Bag of N-grams

- Breaking text into chunks of n contiguous words (or tokens)
- A chunk = an n -gram
- **Bag of words (BoW)** is a special case of the Bag of n -grams having $n = 1$.



Bag of N-grams – Code example

```
from sklearn.feature_extraction.text import CountVectorizer

sample_boN = CountVectorizer(ngram_range = (2, 2))

sample_corpus = ["the cat sat", "the cat sat in the hat", "the cat with the hat"]

sample_boN.fit(sample_corpus)

def get_boN_representation(text):
    return sample_boN.transform(text)

print(f"Bigram Vocabulary mapping for given sample corpus : \n {sample_boN.vocabulary_}")
print("\nBag of 2-gram (bigram) Representation of sentence 'the cat cat sat in the hat'")
print(get_boN_representation(["the cat cat sat in the hat"]).toarray())
```

Bag of N-grams – Code example

Bigram Vocabulary mapping for given sample corpus :

```
{'the cat': 4, 'cat sat': 0, 'sat in': 3, 'in the': 2, 'the hat': 5, 'cat with': 1, 'with the': 6}
```

Bag of 2-gram (bigram) Representation of sentence 'the cat cat sat in the hat'

```
[[1 0 1 1 1 1 0]]
```

Advantages and Disadvantages

- It captures some context and word-order information in the form of n-grams.
- Documents having the same n-grams will have their vectors closer to each other in Euclidean space.
- As n increases, dimensionality (and therefore sparsity) only increases rapidly.
- It still provides no way to address the **out of vocabulary(OOV)** problem.

TF-IDF

- If a word **w** appears **many times** in a sentence **S1** but does not occur much in the rest of the Sentences **Sn** in the corpus, then the word **w** must be of great **importance to the Sentence S1**
- **TF: Term Frequency**
- **IDF: Inverse-Document Frequency**

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

$tf_{i,j}$ = number of occurrences of i in j
 df_i = number of documents containing i
 N = total number of documents

	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7	Document 8
Term(s) 1	10	0	1	0	0	0	0	2
Term(s) 2	0	2	0	0	0	18	0	2
Term(s) 3	0	0	0	0	0	0	0	2
Term(s) 4	6	0	0	4	6	0	0	0
Term(s) 5	0	0	0	0	0	0	0	2
Term(s) 6	0	0	1	0	0	1	0	0
Term(s) 7	0	1	8	0	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0	0

Word Vector (Passage Vector) →

Document Vector ↗

TF-IDF Example

Sentence A = The Car is Driven on the Road

Sentence B = The Truck is Driven on the highway

Word	TF		IDF	TF*IDF	
	A	B		A	B
The	1/7	1/7	$\log(2/2) = 0$	0	0
Car	1/7	0	$\log(2/1) = 0.3$	0.043	0
Truck	0	1/7	$\log(2/1) = 0.3$	0	0.043
Is	1/7	1/7	$\log(2/2) = 0$	0	0
Driven	1/7	1/7	$\log(2/2) = 0$	0	0
On	1/7	1/7	$\log(2/2) = 0$	0	0
The	1/7	1/7	$\log(2/2) = 0$	0	0
Road	1/7	0	$\log(2/1) = 0.3$	0.043	0
Highway	0	1/7	$\log(2/1) = 0.3$	0	0.043

TF-IDF Code Example

```
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf = TfidfVectorizer()

sample_corpus = ["the cat sat", "the cat sat in the hat", "the cat with the hat"]
tfidf_rep = tfidf.fit_transform(sample_corpus)
print(f"IDF Values for sample corpus : {tfidf.idf_}")

print("TF-IDF Representation for sentence 'the cat sat in the hat' :")
print(tfidf.transform(["the cat sat in the hat"]).toarray())
```

IDF Values for sample corpus : [1. 1.28768207 1.69314718 1.28768207 1.69314718]

TF-IDF Representation for sentence 'the cat sat in the hat' :
[[0.29903422 0.385061 0.50630894 0.385061 0.59806843 0.]]

Advantages and Disadvantages

- TF-IDF vectors to calculate the similarity between two texts using a similarity measure like Euclidean distance or cosine similarity.
- We have a fixed-length encoding for any sentence.
- The feature vectors are high-dimensional representations.
- It did capture a bit of the semantics of the sentence.
- They too cannot handle OOV words.

Window-based Co-occurrence matrix

Dictionary: ['I', 'like', 'enjoy', 'deep', 'learning', 'NLP', 'flying', '.']

- Example corpus:
 - I like deep learning
 - I like NLP
 - I enjoy flying
- Window-size = 1

counts	I	like	enjoy	deep	learning	NLP	flying	.
I	0	2	1	0	0	0	0	0
like	2	0	0	1	0	1	0	0
enjoy	1	0	0	0	0	0	1	0
deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

Window-based Co-occurrence matrix

- Similar words tend to occur together and will have similar contexts
- Co-Occurrence: two words have appeared together in a Context Window.
- Example Corpus:
 - Apple is a fruit.
 - Mango is a fruit.

In the above sentence, Apple and mango tend to have a similar context i.e fruit.

- Very high dimensional!

Summary

- Discrete Vector representation
- Many dimensions needed (curse of dimensionality!)
- Vectors do not reflect semantic similarity
- OOV problem

How to represent a text in a low-dimensional space, dense and better at capturing semantic similarity?

Introduction to NLP – Text Representation

(end of lecture)

ONE LOVE. ONE FUTURE.